

Python Programming Exercises

Collection of algorithms and logic implementations (M01 - M10)

Remove Specific Value from List

M01

```
A = [1, 2, 4, 2, 5, 2, 6]
clean_list = [ ]
for i in range(len(A)):
    if A[i] != 2:
        clean_list = clean_list + [A[i]]
print(clean_list)
```

Find Second Largest Number

M02

```
Nums = [9, 2, 11, 7, 11, 3]
big = -100000
second_big = -100000
for i in range(len(Nums)):
    val = Nums[i]
    if val > big:
        second_big = big
        big = val
    elif val > second_big and val != big:
        second_big = val

print("Result is:", second_big)
```

Left Rotation of List

M03

```
A = [10, 20, 30, 40, 50]
first_item = A[0]
n = len(A)
for i in range(n - 1):
    A[i] = A[i + 1]
A[n - 1] = first_item
print(A)
```

Insert Element at Index

M04

```
A = [1, 2, 3, 4, 5]
k = int(input("Index: "))
A = A + [0]
for i in range(len(A)-1, k, -1):
    A[i] = A[i-1]
A[k] = 99
print(A)
```

Delete Element by Index

M05

```
A = [5, 6, 7, 8, 9, 10]
k = int(input("Index to delete: "))
for i in range(k, len(A) - 1):
    A[i] = A[i + 1]
short_a = [ ]
for i in range(len(A) - 1):
    short_a = short_a + [A[i]]
A = short_a
print(A)
```

Find Longest Word in String

M06

```
Text = "I love coding in python"
words = [ ]
current_word = ""
for i in range(len(Text)):
    if Text[i] == " ":
        words = words + [current_word]
        current_word = ""
    else:
        current_word = current_word + Text[i]
words = words + [current_word]
longest = ""
for i in range(len(words)):
    if len(words[i]) > len(longest):
        longest = words[i]
print("Longest is:", longest)
```

Custom String Joiner

M07

```
Words = ["Apple", "Banana", "Cherry"]
sep = " | "
out = ""
for i in range(len(Words)):
    out = out + Words[i]
    if i != len(Words) - 1:
        out = out + sep
print(out)
```

Extract Unique Elements

M08

```
A = [3, 3, 2, 3, 1, 2, 4, 4]
unique = [ ]
for i in range(len(A)):
    found = False
    for j in range(len(unique)):
        if A[i] == unique[j]:
            found = True
            break
    if found == False:
        unique = unique + [A[i]]
print(unique)
```

Element Frequency Counter

M09

```
A = [1, 1, 1, 2, 2, 3, 4, 4]
already_checked = [ ]
for i in range(len(A)):
    skip = False
    for k in range(len(already_checked)):
        if A[i] == already_checked[k]:
            skip = True
    if skip == False:
        count = 0
        for j in range(len(A)):
            if A[i] == A[j]:
                count = count + 1
        print(A[i], "seen", count, "times")
        already_checked = already_checked + [A[i]]
```

Bubble Sort Algorithm

M10

```
A = [8, 3, 5, 2, 9, 1]
n = len(A)
for i in range(n):
    for j in range(0, n - i - 1):
        if A[j] > A[j+1]:
            box = A[j]
            A[j] = A[j+1]
            A[j+1] = box
print("Sorted:", A)
```